

PERFORMANCE TEST REPORT

PharmaTrack - Customer-Facing API Endpoints

This test execution summary reports the results of performance testing for PharmaTrack's customer-facing e-commerce platform. It delivers the outcomes of load, stress, and scalability validation under simulated traffic using Apache JMeter, excluding backend admin and pharmacist features

Document Information

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1. Executive Summary

Performance testing was conducted using Apache JMeter to evaluate the scalability and stability of PharmaTrack API endpoints under load and stress conditions. Testing revealed that while GET endpoints handle concurrent load reasonably well, POST /api/orders endpoint exhibits significant performance degradation under both load and stress scenarios. Overall API throughput of 5.3-5.9 requests/second is adequate for current expected traffic, but response times exceed acceptable thresholds and error rates are concerning at 28%+.

2. Test Scope

2.1 In Scope

- Performance testing of customer-facing API endpoints (4 endpoints)
- Load Test: 50 concurrent users, 60-second ramp-up, 10-minute duration
- Stress Test: 50 concurrent users, 5-second spike, 5-minute duration
- Metrics: Response time (min, max, average, percentiles), throughput, error rate
- Endpoints tested: GET /api/categories, GET /api/products, POST /api/orders, GET /api/orders/:id

2.2 Out Scope

- UI/Web performance testing (page load time, render time)
- Database performance optimization analysis
- Infrastructure-level testing (server CPU, memory, disk I/O)
- Endurance testing (sustained load > 10 minutes)
- Security performance testing
- Admin/Pharmacist endpoints performance

3. Test Objectives

- Evaluate API response time performance under concurrent load (50 users)
- Assess API stability and error handling during traffic spikes
- Identify performance bottlenecks in customer-facing endpoints
- Measure throughput (requests per second) capacity
- Determine if API meets performance thresholds (response time < 2 seconds, error rate < 5%)
- Provide recommendations for performance optimization

4. Test Approach

4.1 Testing Methodology

Load Test: Simulates realistic user behavior with gradual ramp-up. 50 users added over 60 seconds, sustained for 10 minutes to measure system behavior under normal peak load.

Stress Test: Simulates sudden traffic spike. 50 users added within 5 seconds to test system behavior under unexpected surge load.

4.2 Test Execution Tools

- **Apache JMeter 5.6.3:** Load & stress generation
- **Summary Report:** Overall metrics aggregation
- **Aggregate Report:** Percentile analysis and detailed statistics
- **View Results Tree:** Individual request/response inspection

4.3 Test Configuration

- **Delay between requests:** 1000ms (Load Test), 500ms (Stress Test)
- **Response assertions:** HTTP 200 for GET, HTTP 201 for POST
- **JSON extraction:** Capture order_id for sequential request chaining

4.4 Test Environment

Component	Details
Test Tool	Apache JMeter 5.6.3
Test Machine	MacBook 2015
Operating System	macOS Monterey 12.7
Browser	Google Chrome 148.0.7778.168
Target System	Pharmatrack Website
Test Date	April 11, 2026
Network	WiFi

5. Load Test Results

5.1 Configuration

50 concurrent users, 60-second ramp-up, 10-minute duration, 1000ms delay between requests

Total Samples	Avg Response Time	Min/Max Response	Error Rate	Throughput	Status
3458	7942 ms	13 / 19082 ms	28.22%	5.3 req/sec	FAILED

5.2 Per-Endpoint Breakdown

Label ↓	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received ...	Sent KB/sec	Avg. Bytes
POST Cre...	483	9355	2747	15697	2731.24	82.40%	46.2/min	0.41	0.34	543.5
Load Test ...	376	9019	474	15816	2804.23	77.13%	1.3/sec	0.75	0.55	599.7
Load Test ...	375	7497	67	12283	3061.00	13.60%	1.3/sec	13.00	0.18	10387.0
Load Test ...	367	7837	13	19082	3132.31	7.90%	1.3/sec	2.10	0.21	1703.9
Load Test ...	371	6287	50	11980	3203.57	9.16%	1.3/sec	1.99	0.17	1586.2
GET Prod...	496	8145	2244	15287	2951.85	15.52%	46.5/min	7.67	0.11	10123.8
GET Order...	478	8226	2554	13776	3100.78	9.41%	46.1/min	1.23	0.13	1635.9
GET Cate...	512	6956	1304	12371	3199.26	10.16%	47.6/min	1.17	0.11	1509.5
TOTAL	3458	7942	13	19082	3167.39	28.22%	5.3/sec	18.33	1.18	3520.3

6. Stress Test Results

6.1 Configuration

50 concurrent users, 5-second spike ramp-up, 5-minute duration, 500ms delay between requests

Total Samples	Avg Response Time	Min/Max Response	Error Rate	Throughput	Status
3819	7960 ms	66 / 19004 ms	28.99%	5.9 req/sec	FAILED

6.2 Per-Endpoint Breakdown

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received ...	Sent KB/sec	Avg. Bytes
GET Cate...	952	6839	1310	12078	3284.90	9.14%	1.5/sec	2.21	0.20	1521.6
GET Prod...	940	7900	2155	13269	3018.87	13.51%	1.5/sec	14.91	0.22	10349.2
POST Cre...	926	9108	1964	16000	2718.67	80.24%	1.5/sec	0.77	0.65	537.6
GET Order...	912	8129	2571	19004	3099.47	9.21%	1.5/sec	2.35	0.25	1638.5
Stress Tes...	26	7558	406	11992	3654.77	61.54%	1.5/sec	3.21	0.12	2129.9
Stress Tes...	19	8031	1545	11454	2599.10	63.16%	1.0/sec	5.91	0.06	6021.3
Stress Tes...	22	7380	966	10683	3166.62	100.00%	1.3/sec	2.08	0.31	1682.3
Stress Tes...	22	4752	66	11374	3879.29	72.73%	1.3/sec	3.07	0.06	2411.5
TOTAL	3819	7960	66	19004	3160.86	28.99%	5.9/sec	20.20	1.30	3516.3

7.1 Key Findings

GET endpoints (Categories, Products, Order Details) perform adequately under load with error rates under 16%. Response times average 7-8 seconds, which exceed the target of <2 seconds but are manageable. POST /api/orders exhibits critical performance issues with 80%+ error rate under both load and stress conditions. Average response time of 9.3+ seconds combined with high error rate indicates transaction API timeout issues. Throughput remains consistent at 5.3-5.9 requests/second across both scenarios, suggesting API is hitting a scaling ceiling. Response time degradation from Load to Stress test is minimal (~18ms difference), indicating the system stabilizes under spike load rather than progressive degradation.

8. Performance Assessment

Performance does not meet acceptable thresholds. Critical issues identified in POST /api/orders endpoint. Significant optimization required before production deployment.

Metric	Target	Actual
Response Time (Avg)	< 2000ms	7942ms
Error Rate	< 5%	28.22%
Throughput	> 10 req/sec	5.3 req/sec
GET Endpoints Stability	> 95%	85-91%

9. Recommendations

- **Immediate Priority:** Investigate and optimize POST /api/orders endpoint. Root cause appears to be database transaction timeout under concurrent load. Implement connection pooling optimization, query performance tuning, or asynchronous order processing via job queue.
- **Medium Priority:** Reduce overall API response times below 2 seconds by implementing caching strategies (Redis for category/product data), CDN optimization, and database query optimization.
- **Capacity Planning:** Current throughput of 5.3 req/sec supports approximately 450+ concurrent users under normal distribution (assuming 12 requests per user session). If traffic exceeds this, horizontal scaling or database sharding will be required.
- **Load Testing Frequency:** Repeat performance testing after each significant code change or infrastructure update to prevent regression.

10. Conclusion

PharmaTrack API requires performance optimization before production deployment at scale. While GET endpoints demonstrate acceptable stability, the POST /api/orders endpoint is a critical bottleneck with 80%+ error rates under load. Response times across all endpoints exceed acceptable thresholds. Addressing the POST endpoint performance issue and implementing the recommended optimizations are essential before the system can reliably handle production traffic loads.